

Chapter 10

The Unique Role of Academic Health Systems in Facilitating Innovative UI Partnerships



Michelle C. Swick and Edward A. Jones

10.1 Introduction

The innovative fabric of university–industry (UI) partnerships improves lives for people worldwide. Individual nations each have their own unique approach to innovation, with strengths and weaknesses that are rooted in their cultural norms and values. Many emerging markets have extraordinary capabilities in basic science and technology, and they innovate in those spaces with increasing sophistication. While healthcare innovation continues to depend on technology as a cornerstone, application and clinical research are also vital for success.

The “secret sauce” for healthcare innovation (particularly in the United States) is Applied Medical Innovation (AMI). AMI may be defined as innovation achieved by integrating medicine, natural sciences, physical sciences, computation, and engineering to improve healthcare outcomes. AMI is certainly not unique to the United States, but it excels in the United States because the American values of individualism, entrepreneurship, creativity, and free market breathe life into a vibrant innovation ecosystem. Furthermore, Americans who are deeply invested in healthcare innovation know that, at some point, even the best, current standard therapies will fail their patients. They live in a culture that believes *the healthcare we provide today may be good; it may even be excellent. But it’s not good enough for tomorrow.* Finally, many Americans recognize the value of teaching the skills and expertise necessary to keep AMI alive, passing it to the next generation.

One example of AMI revolutionizing healthcare can be found in surgery (Gawande, 2012). In just the past 200 years, surgery has grown exponentially as a viable approach to myriad health concerns. In the 1800s, surgery was limited to

M. C. Swick · E. A. Jones (✉)
Houston Methodist Research Institute, Houston, TX, USA
e-mail: mcswick@houstonmethodist.org; ejones@houstonmethodist.org

managing conditions that were visible from the outside (e.g., amputations) or blood-letting to control inflammation. Procedures were conducted without anesthetics, pain management, or sterility. The advent of anesthesia and antisepsis changed everything. These new concepts allowed surgeons to attempt more complicated, invasive, and precise maneuvers than they had ever imagined. But anesthesia and antisepsis first had to be explored, tested, and implemented through application in the operating room, a place that grows ever more sophisticated with each innovation.

For cardiovascular care, a common thread woven throughout this chapter, the innovations of anesthesia and antisepsis made the heart—a previously “out-of-reach” organ—surgically accessible. After these breakthroughs, cardiovascular surgeries evolved quickly from the first open-heart valvuloplasty for mitral stenosis in 1948 (Harken et al., 1948) to the advent of catheter surgeries for aortic valve implantation in the 1980s to 1990s (Andersen, 2021) to robotic surgeries that achieve unprecedented precision and minimize recovery time. Today, AMI is driving yet another active reorganization of cardiovascular care into an emerging discipline called “structural heart disease intervention” — a new discipline where state-of-the-art techniques from interventional cardiology, thoracic surgery, and radiology disciplines are combined to address anatomical cardiovascular conditions using a holistic and comprehensive approach and unparalleled accuracy (Cubeddu et al., 2009; Raphael et al., 2017).

10.1.1 The Cycle of Medical Translation Drives AMI

How does AMI translate into improved healthcare? It starts with the physician-scientist: a physician who also conducts research in a laboratory, clinic, university, or research institute. This healthcare provider typically possesses both medical doctorate (MD) and Doctor of Philosophy (PhD) degrees. To solve the medical challenges they observe in practice, physician-scientists must become adept at several key skills. First, physician-scientists have a mindset of continuous innovation. They must learn to be identifiers of medical problems and shortcomings. As healthcare providers, they are uniquely positioned on the frontline of current clinical challenges. Thus, they must not only detect where a frustration or pain point exists but also possess a fundamental drive to solve it.

Second, physician-scientists must learn to be dependable, dedicated, translational partners for basic scientists—individuals who can contribute by validating discoveries in real-world settings. Physician-scientists must learn to be speakers of both health system and industry vernacular, using the necessary vocabulary to become a liaison between academia, industry, and other key partners. In this way, they become a biomedical example of the mission of the University Industry Demonstration Partnership (UIDP).

Third, physician-scientists must possess and utilize group-oriented skills and behaviors so they can work together with key partners. These partners include both institutions (i.e., universities, industries, other academic institutes) and individuals

(i.e., engineers, inventors). Each type of institution and each set of experts within it approaches the problem with a different frame of reference. In order to solve the problem, all perspectives must be heard and understood. Such a skill requires specific training and a supportive work culture. As a result, some organizations—academic health systems in particular—must prioritize training and supporting physician-scientists, dedicating the time and resources needed to teach them to be solutions-based thinkers and team scientists.

With these qualities, physician-scientists become part of an essential feedback loop that drives AMI and requires the expertise and synergy of biomedical UI partnerships—the cycle of medical translation (see Fig. 10.1). The cycle begins with a newly identified clinical need or challenge to be solved. Physician-scientists pinpoint the problem and work with their research team to conduct discovery research and uncover a potential solution. The discoveries lead to an invention that must be further developed through applied research. The invention then goes through further product development and preclinical validation. If it proves effective, it can be moved into clinical trials. If successful in clinical trials, the invention is then

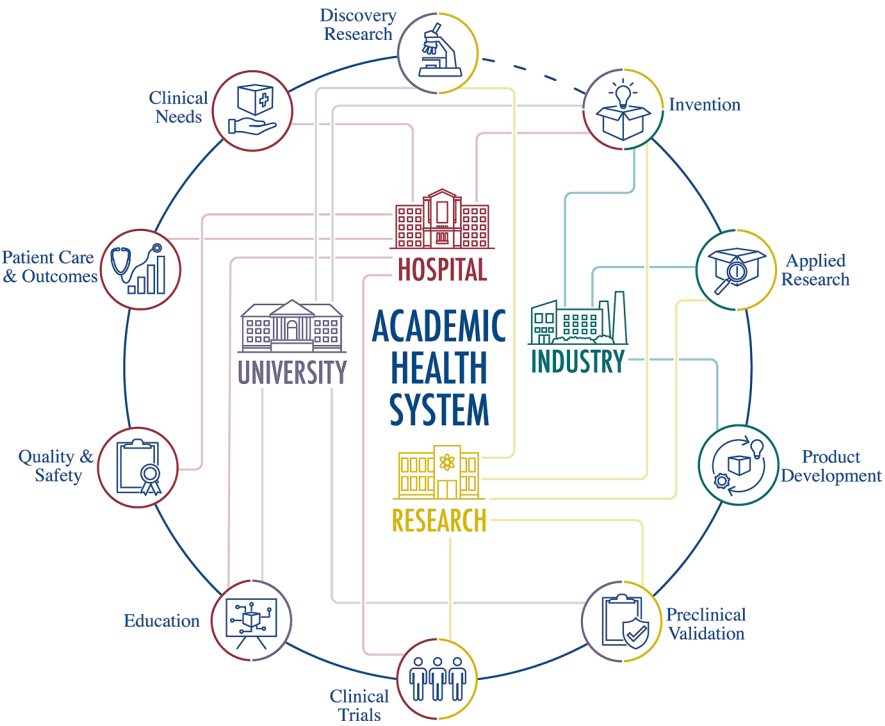


Fig. 10.1 The cycle of medical translation drives AMI. The challenges in translating an innovation into clinical care can only be solved through effective UI partnerships among those who possess the strengths to succeed at each step

implemented through piloting and provider education, evaluated for quality and safety, and scaled for implementation in patient care settings.

Implicit within this cycle, and crucial to a successful journey around it, are two vital elements: successful UI partnerships and physician-scientists. Medical challenges are found in clinical spaces (i.e., the hospital partner), in both ambulatory and inpatient settings. Research at all stages occurs at universities and research institutes, often in collaboration to solve complex problems. Inventions (or innovations) may occur at any partner institution, but product development, preclinical validation, and clinical trials are often supported by industry partners and may occur at research institutes and hospitals. Education and dissemination move from research institutes and universities into hospitals, where patient care is transformed. The physician-scientist with a problem-solving mindset and training in innovation and translational science is essential to catalyzing the cycle, and the cycle is essential to keep AMI moving forward.

10.1.2 Team Science Drives AMI

Nobel Prizes are reserved for only the most impactful, significant, and disruptive discoveries of the times, discoveries that lead to true change in their field. The Nobel Prize in Physiology or Medicine is no exception and has been awarded to discoveries as fundamentally transformative as regulatory RNAs (2006, 2024), viruses that drive cancer progression (2008), and even in vitro fertilization (2010). These discoveries led to such AMIs as RNA vaccines, vaccines that reduce the risk of cervical and liver cancers, and solutions to infertility (*All Nobel Prizes in Physiology or Medicine*, [NobelPrize.org](https://www.nobelprize.org)).

Today, the Nobel laureates and recipients in Physiology or Medicine represent much larger scientific collaborations. They are increasingly representatives and leaders of large, interdisciplinary teams, sometimes numbering in the hundreds. The nature of team science today conflicts with the historical nature of the Nobel Prize, which is more reflective of how science was conducted 100 years ago. One example of this can be found in the words of one of the 2024 recipients, Victor Ambros, who described the discovery of microRNA as a team effort (Ware, 2024):

You can say [the team members are] really the heroes behind this, and our job – mine and Gary [Ruvkun]’s – is to stand in as representatives of the whole enterprise of science, which is so, so dependent upon teams, collaborations, brainstorming amongst multiple people, communications of ideas, and crucial data. (Victor Ambros)

Since the 1950s, there has been an increasing tendency to award the prize to more than one person. There are several reasons for this. First, the problems being solved are increasingly complex and often demand interdisciplinary teams and partners to solve. Second, increases in scientific collaboration lead to two or three leading minds working together to make the discovery. Conversely, competition within a field can lead to two or more people arriving at a solution at nearly the same time.

Thus, despite being competitors, two or more individuals may receive the prize together though they did not work together on the idea.

Since 1990, one in every two prizes has included at least one physician or physician-scientist recipient. Why might this be the case? We posit that one reason may be because the well-trained physician or physician-scientist is the front-line detector of clinical challenges in need of AMI. A physician with an innovative mindset and culture can collaborate, invent, and truly solve a problem with creative, disruptive impact. While we cannot confidently confirm whether any given MD-credentialed Nobel Prize recipient collaborated with their co-recipients without extensively greater investigation of each discovery's history, we postulate that, at least in some cases, the contribution of the practicing physician or physician-scientist is essential for significant, applicable impact on patient care. These thinkers are acutely aware of where the problems lie. They are the boots on the ground, facing the challenges directly.

10.1.3 Culture of Innovation Drives AMI

Where does AMI happen? An Academic Medical Center (AMC) is traditionally defined as a university or medical school and its partner teaching hospital. The combined assets include the teaching hospital, the clinical faculty who are jointly appointed to the university, and any affiliated private-practice physicians who practice in the community (non-academic). This is often an accurate definition for any given AMC, though organizational structures can vary widely. By contrast, a Community Medical Center (CMC) is traditionally a nonacademic healthcare provider, and this term is often synonymous with a community hospital.

The terms “academic medical center” and “academic health system” are frequently used interchangeably. However, for the purposes of this chapter, we wish to distinguish between an AMC and an academic health system. An academic health system is not exclusively academic—like a university with its partner teaching hospital—nor is it an industry partner. Like any hospital, the academic health system's top priority is the patient population. Unlike a hospital, the academic health system's priorities after patient care and outcomes are a mix of academic and industry interests. Academic health systems are deeply invested in innovations that improve patient care or transform the patient experience.

The Houston Methodist Health System (HM) considers itself an academic health system. A handful of the highest-quality, broad-spectrum, tertiary care health systems in the United States, including Cedars-Sinai Medical Center, Cleveland Clinic and the Lerner Research Institute, MedStar Health, and Salt Lake Clinic—Intermountain Health, among others, fit this academic health system model well. Of course, each will have distinct variations in structure, but they share a common goal—to translate their AMI into solutions they can deploy across their system and beyond. Here, we will use HM as an example to illustrate the academic health system as it is the one with which we are most familiar (see Fig. 10.2).

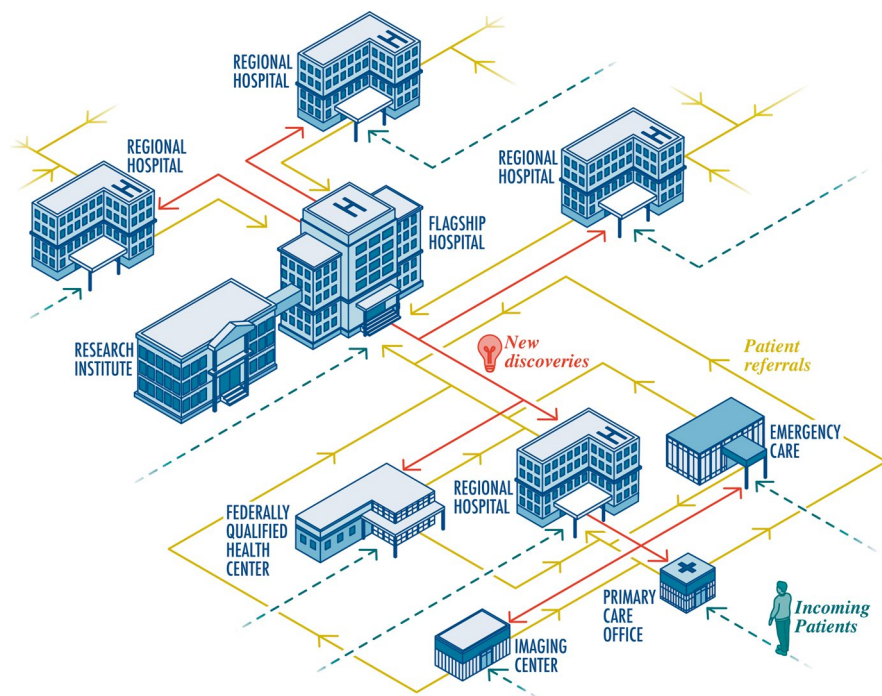


Fig. 10.2 Structure of an academic health system. Incoming patients (dashed lines) access the care network at any point. Patients may then be referred across the system (green solid lines) to the location best suited to meet their current needs. Discoveries and innovations made at the research institute through UI partnerships are first tested in the flagship hospital and then pushed (red solid lines) to regional hospitals, FQHCs, and other locations where they can reach patients close to their homes

HM is anchored by an AMC flagship location consisting of Houston Methodist Hospital and Houston Methodist Research Institute. This flagship is supported by eight regional hospitals, each with a robust open medical staff of employed and private-practice physicians. These regional hospitals are frequently mislabeled as CMCs. In reality, research and teaching take place in many of these hospitals as well, amplifying the clinical and research enterprises of HM in a “hub-and-spoke” model. Beyond these hospitals are Houston Methodist primary care offices, emergency care, and imaging and physical rehabilitation centers, all of which serve as direct entry points for patients to access the whole system.

An academic health system must find innovative ways to serve the community at large, not only their consumers/patients. This may be through direct patient interactions or through strategic partnerships with community clinics. Thus, beyond its hospitals, medical offices, and imaging centers, HM has partnerships with community-based, federally qualified health centers (FQHC). FQHCs seek to address the unmet needs of those who find themselves lacking access to certain healthcare services. Partner FQHCs have different health delivery models and

distinct, often underserved, patient populations. HM works to disseminate innovations to these partners too, validating innovations in these different settings. This effort creates an opportunity to discover the needs of a broader patient population and identify gaps in care. It also serves as a chance to discover and overcome the challenges of scaling AMI to serve in a wider range of environments and healthcare delivery models.

The result of this organizational structure is a robust, bidirectional academic pipeline of entry points for patients to gain access to clinical trials. HM takes the research to patients rather than the other way around. Academic health systems like HM provide a culture of innovation and a teaching infrastructure that drives AMI. They prioritize physician education in translational science and in new technologies and techniques for broad patient use. Beyond the players illustrated in Fig. 10.2, HM is purposefully pluralistic in its partnerships, each one formed with the intent to augment its expertise using a best-of-breed approach. Because of their unique infrastructure and key resources, they are ideal places to train physicians in the newest techniques that come to market, improving the efficiency of medical translation and solution adoption.

Because of its unique composition, HM and academic health systems like it have a key role to play in UI partnerships as facilitators. An academic health system is an exclusive player in this space because it brings all dimensions together—science, medicine, teaching, and innovation—and they have unique capabilities to achieve unparalleled excellence. Next, we will highlight some of the unique advantages of working with academic health systems, while also discussing the reasons behind some of the challenges partners can face when working with an academic health system.

10.2 Opportunities and Challenges

An academic health system is a unique organizational vehicle for UI partnerships that focus on AMI. Such a vehicle can facilitate the research and development activities necessary to advance both technology and knowledge from concept to application. For example, academic health systems have the ability to hire different types of personnel, such as scientists, engineers, and technicians, who can take on roles that are considered atypical or even unfeasible in a traditional hospital. They can also create different types of facilities, equipment, and data security to achieve innovation while protecting patient data. Furthermore, academic health systems have access to a diverse patient base and the capacity to deploy biomedical solutions where patients live. Finally, the ideas crafted in the academic health system environment have already undergone a rigorous test of their application in a real-world setting, and are therefore scalable for broad adoption. In contrast, this implementation step is one of the challenges that AMCs face while deploying their innovations; technology availability often limits broader real-world application.

Academic health systems have other process-based advantages as well. For example, they have the capability to facilitate different types of intellectual property through their office of technology transfer. Innovation is strategically prioritized, giving the academic health system the ability to establish an infrastructure, which may be supported by philanthropy, that bridges the “valley of death” in the cycle of medical translation. Such unique infrastructure might include portfolio project managers, intramural research funding to advance the project, and critical equipment, facilities, and expertise to meet in-human validation and scale-up requirements.

Likewise, an academic health system has some disadvantages as a UI partner, particularly when compared with other entities that do not serve patients. Academic health systems must operate within the same regulatory limits as any hospital. Hospitals and academic health systems are subject to all fraud and abuse laws set forth by the Office of Inspector General (*Fraud & Abuse Laws*, oig.hhs.gov). According to these laws and regulations, established by Congress, hospitals and academic health systems must operate within the rules of Medicare to be eligible for reimbursement for health services rendered. These include the Anti-Kickback Statute [42 U.S.C. § 1320a-7b(b)] and the Physician Self-Referral Law, or “Stark” law [42 U.S.C. § 1395nn]. Simply put, these laws prohibit physicians from conducting unnecessary procedures or encouraging inappropriate or biased referrals in exchange for a reward. These laws protect patient safety first and foremost, as well as hospital integrity. However, they can also prevent a partnership agreement from taking a more familiar or “typical” appearance that might be legal and ethical in other sectors. Government regulations also drive hospitals and academic health systems to align their priorities with those of the government, specifically, ensuring that Medicare and Medicaid beneficiaries receive the highest possible value. This can sometimes cause the academic health system to be perceived as a difficult partner; however, operating within these critical laws and regulations is essential to protect the academic health system’s top priority—the patient.

10.2.1 Mindsets and Motivations

Generally, leaders and funders of biomedical research are shifting their policies and mindsets away from siloed, basic research alone and toward integrated team science, with portfolios of investments that include both applied/translational research and mission/challenge-driven research. In other words, funders are not as interested in “knowledge for knowledge’s sake” but rather knowledge with powerful impact on patient outcomes, the patient experience, or both—knowledge that can drive AMI. Academic health systems have this same mindset.

When viewed through a patient outcomes lens, it is easy to see why a health system would be motivated to pursue an academic portfolio with UI partnerships. The resulting advancements from invention, validation, and implementation of a new innovation will improve patient outcomes, patient experience, or both. The hospital’s mindset is rooted in the Latin words of the original Hippocratic

oath— *Primum non nocere* (First, do no harm). Hospitals and their employees are acutely aware of the trust their patients have placed in them and of the vulnerable nature of patients facing medical challenges. They acknowledge that the currently available therapies and procedures may not be effective or curative for everyone. Thus, they will approach exploring new solutions with patient safety in mind, but they are highly motivated to find solutions to health challenges that can be game-changing for patients.

Conversely, when viewed through a strictly business lens, it is difficult to see why a health system would want to pursue this same academic portfolio with UI partnerships, particularly in the current healthcare and economic environments. It is expensive and has a high regulatory burden. In this sense, a clinic or other nonacademic hospital represents a simpler, more formulaic mission: to provide excellent patient care at the current best standard of practice. The cost to innovate and deliver a new health solution often negates profit; it may even lose money in the process. This is why HM and others like it are nonprofit and mission-based. They do this work as a service to improve the health of patients.

10.2.2 Challenge: Cost to Innovate

Hospitals may operate under for-profit or nonprofit models, but in any case, much of their revenue is generated through private and public/government (i.e., Medicare and Medicaid) healthcare provider agreements. These agreements are meticulously negotiated between the financial teams of hospitals and insurance companies and set agreed-upon prices for every medical service the hospital provides. These prices must exceed expenses for the hospital to maintain a positive financial margin. Unfortunately, margins run notoriously thin, and costs to deliver healthcare are rapidly increasing, particularly because of the post-pandemic inflation that the United States, and the world, have experienced from 2022 onward.

The data describing health system margins in the past five years have been heavily affected by the COVID-19 pandemic, which clouds our ability to assess “typical” margins using current data. With that caveat, margin analysis in 2023 showed a wide spread of hospital margins, from -6.8% to 12.2% , which is an improvement from 2022 (Ashley & Condon, 2024). This range of margins, spanning from loss to net-neutral to modest gain, is wider in the wake of COVID-19 than in years past. These data indicate that many hospitals operate with thin-to-zero margins, which means there are times when they must delay or forgo investments in new technologies that directly benefit patients. If break-even margins are not achieved, the hospital ceases to exist, a problem that is growing increasingly pervasive, particularly in rural areas. And yet, hospitals must invest in new technology if they hope to enhance healthcare delivery, improve patient outcomes, and compete with other systems.

10.2.3 Challenge: Regulatory Burden

Hospitals often hesitate to engage in clinical research because of the extensive, complex regulatory burden that the process entails. Research hospitals must comply with all required federal, state, and local regulations and policies, such as those set forth by the Food and Drug Administration, the Department of Health and Human Services, and local Institutional Review Boards. Navigating and fulfilling these requirements demands substantial administrative resources, specialized legal expertise, and constant updates on shifting policies, which can divert focus from a hospital's primary mission of patient care. Additionally, compliance failures can result in significant penalties or legal consequences, further discouraging hospitals from involvement in research.

The intricacies of managing patient data present another significant hurdle. Hospitals must adhere to strict data protection regulations under the Health Insurance Portability and Accountability Act of 1996 (HIPAA), which safeguards patient information in a highly regulated environment. To be HIPAA-compliant, hospitals require specialized data infrastructure, increased staff training, and the adoption of advanced technologies to handle and secure data appropriately. This comes at a high cost and requires ongoing investment, which many hospitals—especially those with limited resources—find challenging to maintain. When one attempts to introduce a research use for these sensitive data, the complexity increases further. Researchers must obtain informed consent from patients, and studies involving vulnerable populations (e.g., children, pregnant persons) or critically ill patients have yet another layer of difficulty and demand careful regulatory oversight.

10.2.4 Challenge: Problem Complexity

Today's medical problems are often too complex to be wholly owned and solved by a single institution or even a single type of institution. Historically, this has always “felt” true. A problem without an answer feels complex until it is solved. For example, preventing infection was considered unsolvable by most until Louis Pasteur first articulated germ theory, inspiring the works of Joseph Lister and Robert Koch (Lepenos et al., 2024). However, today's challenges have an added complicating factor: the state of technology. Technology drives innovation and solves problems at a faster speed than ever. It also creates new problems, such as data security and unintentional bias.

If a single institution attempts to innovate alone, it is likely limited to incremental advancements. Transformation is going to require the convergence of people from different backgrounds, different disciplines, and different training. This will require academically oriented institutions, including both universities and academic health systems, to enter partnerships that may feel traditionally nonobvious but are essential in the face of technology factoring into the solutions. Thus, academic

health systems should frequently and enthusiastically partner with biomedical companies, such as Medtronic, Pfizer, J&J, and AstraZeneca. They should also consider adding other organizations to their portfolios, such as technology experts at Microsoft or Vidya and engineering/manufacturing experts at GE or Siemens. Incorporating less-obvious types of computational and engineering partnerships will be increasingly more important going forward, as recognized, for example, by some of the National Science Foundation's newest award mechanisms and calls for proposals (*Technology, Innovation and Partnerships*, [new.nsf.gov](https://www.nsf.gov)).

10.2.5 Challenge: Super-Sub-Specialization

There are two types of experts: the generalist, who has breadth of knowledge, and the specialist, who possesses a depth of expertise. Super-sub-specialization is desirable for many reasons. First, patients want and feel safe with a specialist who understands their situation deeply. Second, the super-sub-specialist is someone to rely on to be correct and highly proficient in a complex topic or skill. However, the ideal innovating partner has T-shaped skills—both breadth and depth of expertise (see Fig. 10.3). Unfortunately, many of today's experts are so super-sub-specialized that they cannot see the big picture and thus cannot solve complex problems. They can skillfully apply their knowledge to an innovation, but they do not innovate. With training in AMI, however, the super-sub-specialist can learn to engage with partners and drive AMI. Thus, the more sub-specialized we get, the more we need to reach

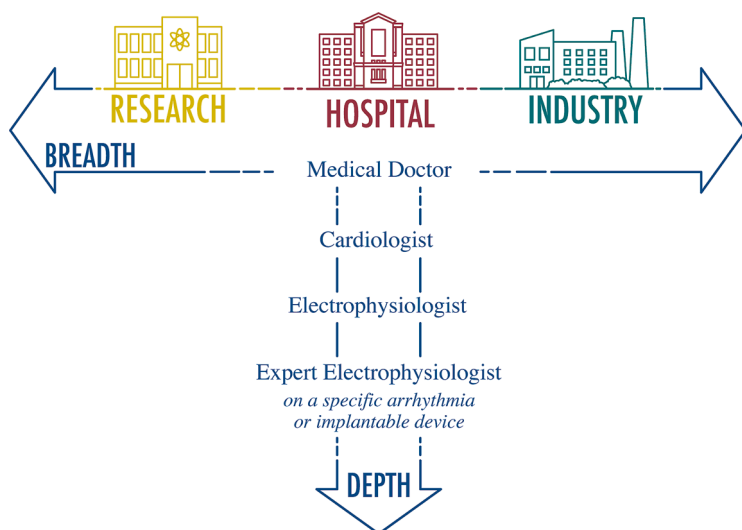


Fig 10.3 T-shaped skills in applied medical innovation. Successful AMI requires physician-scientists with T-shaped skills: depth of expertise in a specific area to identify challenges and breadth to communicate with partners

out to new partners. We need a myopic lens to find the problem, but then a wide-angle lens to see the big picture and view our partners. To break out of a “hive mind,” specialists can surround themselves with people who think differently. Innovations from teams of like-minded people will be incremental at best. It is only when people who think differently come together that innovation happens.

10.2.6 Challenge: Pace of Change, Particularly of Technology

Technology drives innovation, as it always has. Data are simultaneously the fuel and the product of technological innovation. Data increasingly solves existing problems while creating new ones and serves as a source of new insights. As new technologies come online, they create new, faster, more efficient ways of doing things, which leads to new solutions—and new problems.

Technology also advances at lightning speed. In 1965, the co-founder of Intel, Gordon E. Moore, predicted that the number of components in a single computer chip would double every 2 years (Moore, 1965). This insight became known as “Moore’s Law,” and it has held up surprisingly well over nearly 60 years. In the age of deep machine learning and large-scale data, computing power is now doubling every 6 months (Sevilla et al., 2022) and reshaping what is possible for organizations and UI partnerships between them. It is therefore expensive, but necessary, to invest in technology in order to remain on the leading edge of innovation.

Technology also creates new complexity and new problems that require partnership to solve. Furthermore, the solutions to these complex problems must occur securely, in ways that are appropriate for the setting. For example, an interesting solution might be effective in the context of conducting research on nonhuman specimens, but not practical for humans in a medical setting. If an engineer designs a new device or tool to solve a medical problem but has never set foot in an operating room, the tool may prove impractical, unusable, or even unsafe. Thus, the partnership of expertise is paramount to our ability to anticipate and innovate through all possible obstacles.

10.2.7 Challenge: Physician-Scientist’s Responsibilities

The physician-scientist experiences an extraordinarily difficult strain on their professional time to meet the expectations of their patients, employer, mentors, mentees, and all governing bodies from their organization to the federal government. Clinic time, continuing medical education requirements, student mentoring, personal career development/promotion, and pursuit of research funding leave very little time for research projects. As if to add insult to injury, investigators have reported that they spend nearly half (44.3%) of the time available for their research projects on administrative tasks, including regulatory compliance (Schneider,

2020). Regulatory compliance includes periodic (usually annual) certifications for biosafety, HIPAA compliance, information technology (IT) security and safety, clinical and animal research practices, conflicts of interest, and human resources. However essential it is for safe and responsible conduct of research, regulatory compliance places an enormous time burden on physician-scientists, who are described as the “point of integration” in the compliance pipeline (Carmichael, 2024).

10.2.8 *Challenge: Spectrum of Partnerships*

As a participant in UI partnerships, engagements with academic health systems like HM can vary and lie on a spectrum from transactional to innovative (Fig. 10.4). A partnership that may be labeled as “transactional” is one where an innovation or invention is brought to the table by another partner. For example, if a partner has developed a new drug for triple-negative breast cancer, HM, as home to a leading authority in that cancer, may be an ideal location for a clinical trial. HM’s priority and motivation in this partnership are to be a catalyst for patient care improvements, confirm value added if it exists, and deploy rapidly to patients who will benefit. By contrast, other UI partnerships wherein an academic health system participates may be more innovative. For example, one of HM’s faculty researchers has an innovation and a patent on the idea but needs a partnership to support scale-up. While patient care improvement is still top of mind, the academic health system also focuses on protecting the intellectual property. HM approaches every new partnership as innovative, in case some intellectual property emerges, ensuring that protections are in place for that contribution, so that the investments of time, money, and space are rewarded.

Perhaps unsurprisingly, UI partnerships are the only path through these myriad challenges. UI partnership brings the thought innovators and the problem solvers into the same space, creating an opportunity to move ideas forward. Academic health systems, in their role as facilitator, can augment and magnify the impact of these partnerships to accelerate innovation from discovery to care delivery.

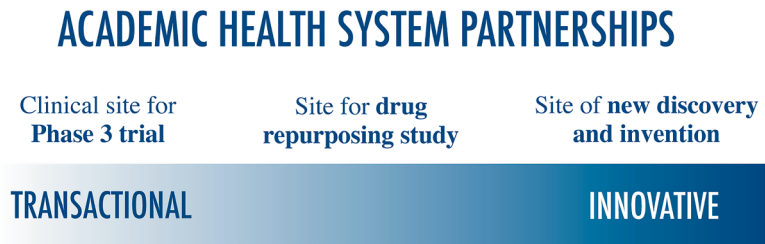


Fig. 10.4 A spectrum of partnerships. An academic health system may be a smaller partner, e.g., generating real-world data for a clinical trial (transactional). It could also be the site where the inventor works, and thus become more deeply involved in the full cycle (innovative)

10.3 Prior Research

Many terrific frameworks have been presented by UIDP partners and others to guide how UI partnerships should be approached and designed. These frameworks and models can enlighten members of each organizational type, revealing glimpses into the mindset and perspectives of their partners. Such models instill value by helping each partner to recognize and respect the differences in priorities that not only make the partnership challenging but also make it worthwhile, rewarding, and, for health-care innovations, life-saving. Here, we will briefly reflect on a few relevant frameworks through the lens of the academic health system.

The Partnership Continuum Framework was described by UIDP in 2012 as a means to capture the range of complexity of UI partnerships and roughly categorize them based on the common themes that emerge from each type (Umberger et al., 2012). In this model, partnerships are described on a continuum with levels of engagement, from a transaction (shared tactics), to a collaboration (shared ideas), to an alliance (shared aspirations). The model can be applied to partnerships with academic health systems much like it is for universities and AMCs. Academic health systems engage in partnerships from Level 1 (transaction) to Level 3 (alliance), may enter an initial partnership with a new partner at any level, and may have multiple partnerships with the same partner at all levels simultaneously.

The Triple Helix Model for UI partnerships incorporates government relations as a third partner, hence forming a “triple helix” (Etzkowitz & Leydesdorff, 1995). For academic health systems, the government can be a partner; however, government entities function more as the operating bodies that codify and set forth regulations and standards. Thus, academic health systems typically view the government as more of a regulating body than a partner, with notable exceptions. In cases where research is funded by a government agency under competitive contract negotiations, the government forms a more partnership-like arrangement. In another adaptation of the Triple Helix Model, the academic health system could be contemplated as the third partner in the helix, with some modifications.

The King–Persily Model for UI partnerships attempts to solve one key issue between universities and industries when navigating process and ownership surrounding data and proprietary information. This model introduces a third-party commission to decouple full access to industry data, policies, and proprietary information from the scholars/researchers conducting the studies (King & Persily, 2019). Initially developed for social sciences and social media big data, this model may have some applicability within the biomedical research innovation space, but with modifications. First, academic health systems tend to reserve the right to publish but agree to offer industry partners a window of time to review and redact proprietary information. Second, all patient data are deidentified to make it usable while protecting the privacy of patients and research participants.

We note that patient data presents an ethical conundrum, not only when considering a given UI partnership framework, but in every partnership an academic health system forms independent of which model the partners follow. Patient data

represents one of the academic health system's major value propositions within their UI partnerships. As such, patient data holds enormous value and cannot simply be given away to a partner, even if deidentified for privacy protection. People who don't quite know how to interpret the data may arrive at the wrong conclusions, and incorrect conclusions can cost lives. Patient data also should not be monetized solely for financial gain, in part for the same reason as above but also because academic health systems have been entrusted by the patients and participants in the trials to protect their health information and use it judiciously, ethically, and for purposes consistent with the informed consent they provided.

10.4 Options, Solutions, and Impact

Partnerships between industries, universities, and academic health systems have significant impacts, but also face considerable challenges. Possible solutions to improve partnerships between universities and industries are frequently contemplated but remain elusive. Leaders from both organizational vehicles have long recognized the need for a shared vision and the challenge of obtaining buy-in from partners with very different goals and priorities. Recent experiences in healthcare might reveal some curious lessons learned.

10.4.1 *The COVID Experience*

Each member of a UI partnership enters a new engagement with certain priorities and concerns, both of which are usually driven by their priority stakeholders and are entirely valid. The concerns drive a wedge between the problem and the solution. But could solutions to improve UI partnerships be hiding in a reflection on the COVID-19 pandemic? Could the COVID-19 experience make us question how to face risk? Might academic health systems bridge the gaps in some of these partnerships?

The COVID-19 pandemic was a fraught and agonizing time. Leaders at organizations of all kinds adopted the same mindset: "whatever it takes to get an answer, do it. Learn from that answer, adjust quickly, and try again." This highly iterative process was very efficient. Everybody could see the risk of not acting—unnecessary loss of life and prolonged economic damage. The strain of this monumental challenge revealed many strengths. Under the duress of the pandemic, industry and academia chose to set aside differences in order to get things done. This decision, combined with ample funding and expectations outlined in Operation Warp Speed, produced multiple, highly effective vaccines in record time (Slaoui & Hepburn, 2020). AMCs and academic health systems across the country united as sites for Phase 3 trials in partnership with industry leaders Pfizer, Moderna, and Johnson &

Johnson. Vaccinology advanced in 1–2 years to a level that would otherwise have taken decades.

COVID-19 also revealed some weaknesses in UI partnerships. The increased cooperation was short-lived. In the post-COVID-19 era, UI partners have very quickly returned to hitting the roadblocks that counter currents form. Partners resumed old habits. In some cases, the pendulum swung back to an even more conservative approach. While the actions and choices made by UI partners in direct response to COVID-19 are not sustainable as a permanent model, they do illuminate what can be achieved when partners have aligned goals and a strong sense of urgency.

Why is urgency so subjective with regard to human disease? The answer may come from the inherently different forms that diseases can take. COVID-19 was an indiscriminate, highly communicable disease, which made the sense of urgency very easy to feel. This contagious, mysterious threat made the need to act at any cost obvious. By contrast, cardiovascular disease, cancer, and neurodegenerative conditions are noncommunicable diseases (NCDs); people do not worry about “catching” them from the person seated next to them in a theater or on an airplane. They don’t need to worry they might bring them home to an at-risk loved one. NCDs tend to have a slow progression, and thus they give the appearance of being deferrable.

But just like COVID-19, NCDs are also harming enormous numbers of people. For a patient with an NCD, and for their caregivers, their situation is just as urgent as COVID-19 was to the global population. Unfortunately, partners with the power to be change agents may also lack any direct experience with the harms of the NCD on which they are working. They also likely have stakeholders to answer to. When a disease progresses slowly over years, it is not so easy to see the impacts in real time. However, academic health systems especially are tuned into one undeniable fact—at some point, standard therapies fail patients with NCDs as well.

COVID-19 has illuminated much about UI partnerships that have AMI as their focus. Health systems can assume risk. Industries can assume risk. Patients want to participate in trials and be part of the solution to health challenges. Healthcare workers are capable professionals who can rise to challenges and deploy modified care models to meet the moment. And all of this could be deployed quickly when motivation and urgency are high. If partners can fully appreciate the sense of urgency for those facing serious, life-threatening NCDs—by placing patients at the center—it could be a motivating factor for partners to approach their UI partnerships. Academic health systems are ideal facilitators to help keep the focus on the patients.

10.4.2 A Values Alignment Model

How do UI partners align themselves around common goals for AMI while respecting the priority differences of their partners? Here, we contemplate a *values alignment model* wherein the academic health system serves as a facilitator that pulls all

partners toward the values they share while respecting the values they don't (see Fig. 10.5).

For every partnership we enter, we must ask ourselves: What does “values alignment” mean? What counter currents are pulling us away from alignment, and how can we reconcile those forces to realign our goals, balance the model, and achieve innovation together? In biomedical innovation, the aligned goal usually starts with, “We want to treat X disease.” The hospital prioritizes patient safety and privacy. The industry partner contemplates the potential market for the product and searches for a means to obtain validating, real-world data. The university partner advocates for research, development, and student training. The physician prioritizes not only patient care, but relatedly their own reputation, independence, and publication record.

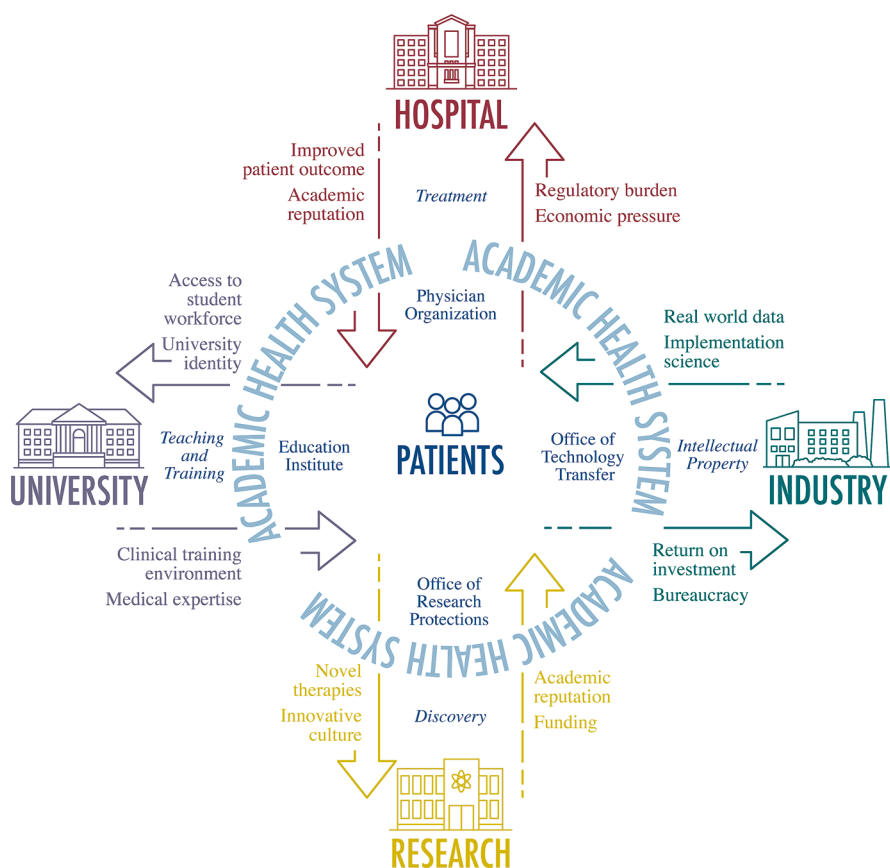


Fig. 10.5 A values alignment approach to UI partnerships. Academic health systems place patients at the center to drive AMI. The academic health system can then facilitate patient protection while balancing the push and pull from uniting priorities (arrows in) and countercurrents (arrows out)

Just as each partner's priorities differ, so too does their definition of return on investment (ROI). For an academic health system, the greatest ROI is providing the highest quality patient care. This leads to a strong reputation, where patients choose to return to the academic health system because of the options available to them. For industry, the team is looking for a product to market with maximal profit, but this is also dependent on improved patient experience and less patient suffering. University partners are searching for environments and circumstances that create an ideal competency that is cutting-edge, reliable, relevant to patients whom their students will one day serve.

The values alignment model we have set forth centers the focus on what priorities partners have in common while acknowledging and respecting the values that they don't by viewing all priorities through a patient-centered lens. The challenge for hospitals and universities is to respect the pressure industry faces with ROI, shareholders, and timelines. We must create products that will earn a profit for industry because they appeal to patients. The challenge for industry is to respect the protective obligations a hospital has for its patients. We must develop, test, and implement new solutions safely, aligned for patient benefit.

10.5 Case Study

The Houston Methodist DeBakey Heart & Vascular Center, long led by its namesake and pioneering heart surgeon Michael E. DeBakey, has been a key site for UI partnerships and several innovations in cardiovascular surgery. One such innovation, and the subject of this chapter's case study, is the Transcatheter Aortic Valve Replacement (TAVR) procedure to treat aortic stenosis. The Methodist DeBakey Heart & Vascular Center was a critical member of a UI partnership to validate the TAVR procedure. As a result of this partnership, treatment for patients with aortic stenosis has been revolutionized.

Aortic stenosis is a type of heart valve disease where the aortic valve, located between the left ventricle and the aorta, narrows and doesn't fully open, thereby restricting blood flow from the heart to the rest of the body. The disease has a long latency period, but once symptoms emerge, they worsen rapidly and cause high mortality rates among untreated patients. If symptomatic and left untreated, average survival is only 2–3 years.

Aortic valve replacement (AVR) is the primary means of treating aortic stenosis and improving outcomes for patients. Historically, this surgical intervention was conducted as an open-heart procedure (surgical AVR, or SAVR). As such, patients with aortic stenosis who had other comorbidities were at higher surgical risk and thus ineligible for such an invasive surgery. Other patients, while eligible for SAVR, had increased surgical risk to a level where they avoided the invasive procedure for various reasons (Bach et al., 2009). Biomedical innovation was the only way to create a solution for patients with aortic stenosis who could not endure an invasive, open-heart procedure.

Physician-scientists at Skejby University Hospital (Denmark) developed the concept of TAVR in the 1990s as a disruptive technology to circumvent the risks of SAVR (Andersen, 2021; Bonhoeffer et al., 2000; Knudsen et al., 1993). The first transcatheter valve was placed in a human by accessing the heart through the femoral artery at Charles Nicolle Hospital, University of Rouen, France, in 2002 (Cribier et al., 2002). Very quickly thereafter, additional cases were published, and technology patents were filed. A contentious patent battle unfolded between Medtronic and Edwards Lifesciences, a topic beyond the scope of this chapter. Importantly, Medtronic's technology had additional innovations that improved patient safety. With the Medtronic device, known as CoreValve, the surgeon can reposition the valve to ensure it is properly seated before securing the device.

Medtronic partnered with the DeBakey Heart & Vascular Center and other clinical sites to conduct clinical trials on the repositionable, self-expanding, transcatheter aortic-valve bioprosthesis. The original study, known as the PARTNER trial, ran from 2007 to 2009 and was subsequently published in the *New England Journal of Medicine* (Leon et al., 2010). The trial included only patients who were considered ineligible for a SAVR procedure due to high risk or other comorbidities. When valve replacement is not an option, the next best standard of care is a valvuloplasty. In the study, the rate of death from any cause was significantly reduced in patients receiving the TAVR procedure compared with those who received the next best standard of care. Patients receiving TAVR also spent significantly less time in recovery and had a lower rate of rehospitalization than their control counterparts.

Quickly, the clinical trials progressed to test TAVR outcomes in additional subsets of patients with aortic stenosis. After the initial trial, which was in SAVR-ineligible patients only, TAVR was compared directly to SAVR in patients eligible for either procedure, first in the most severe cases (Smith et al., 2011), then intermediate cases (Reardon et al., 2017), and then even in mild and moderate cases where SAVR might be postponed because of the intensity and severity of the surgery, not the clinical condition (Popma et al., 2019).

The volume of TAVR cases increased annually from 2011 to 2019, and in 2019 TAVR volume surpassed that of SAVR. Today, TAVR is largely viewed as the standard of care across the entire spectrum of surgical risk for aortic stenosis. It is also expanding into new patient populations (younger, lower-risk) with a wider range of anatomies. It is taught to surgeons across the United States at HM, Cleveland Clinic, Mayo Clinic, and many more and is available to patients in all 50 states (Carroll et al., 2021). At HM, TAVR is offered at the Valve Clinic in six hospitals within the system. The Valve Clinic at Houston Methodist within the Texas Medical Center (flagship hospital) is one of the most experienced TAVR sites in the nation, with over 2000 device implantations from one integrated TAVR team across the Greater Houston area.

As TAVR technology expands, it experiences continuous, iterative innovation and improvement, now referred to as transcatheter heart valve (THV) design. Every iteration gains on the previous one. Now there are balloon-expandable valves (BEVs), such as the Sapien X4 (Edwards), and self-expanding valves (SEVs), such as the Evolut FX (Medtronic). These subsequent innovations have led to lower

30-day mortality (7.2% to 2.5%) and lower stroke risk (2.75% to 2.3%) (Carroll et al., 2021; Zaid et al., 2023).

In 2023, TAVR was projected to be an economically dominant strategy with a 95% probability that the incremental cost-effectiveness ratio for TAVR would be less than \$50,000 per quality-adjusted life-year gained (consistent with high economic value from a US healthcare perspective). TAVR also has many intangible benefits, particularly to the patient—less time in hospital, fewer readmissions, and reduced infections and other complications.

TAVR serves as a genuine example of the benefits of UI partnerships that include an academic health system partner. HM had (and still has) the culture, the ecosystem, the clinicians, and the patient volume/pipeline to do the work. Though industry-oriented questions/concerns, such as the economics and ROI of the investment, are critical, the academic health system partner keeps the focus on “what’s in the best interest of the patient?” Ultimately, if the innovation is not safe and does not significantly improve the patient experience, ROI potential goes down to zero. Academic health systems are extremely well positioned to balance patient outcomes with economic viability.

TAVR has changed the paradigm in how we treat valve repair. This is a case where everybody wins, but the patient remains the most important beneficiary of the investment and effort. Without the partnerships between industry (Medtronic) and the various academic partners (including HM), this innovation would not have been possible. The hospital alone, already running on thin margins, might have decided this endeavor was too expensive or risky. The university alone, without clinical context, might not be able to translate the technology effectively. The industry alone, without sufficient real-world data, might have missed a critical element of patient safety or operating room mechanics that could render the procedure implausible. The scientist alone, without the proper infrastructure and collaborations surrounding them, might have overlooked critical factors that could prevent scalability outside of a single AMC by moving too fast or too far without a critical piece of information only obtained through partnership with trained physician-scientists or engineers. This success would be unlikely without the academic health system as the nexus for the partnership, serving to handle the complexities.

10.6 Takeaways and Future Strategies

Catheter technology was not new at the time of the first TAVR procedure. In fact, catheter use can be traced as far back as 3000 BC to drain the bladder. It was through creative, robust UI partnerships that catheter technology could be leveraged and innovated in this transformative way. TAVR revolutionized treatment for aortic stenosis, and even opened doors to new UI partnerships that seek to tackle other catheter-based cardiovascular surgeries and beyond. Catheter technology is being applied in both related and fully novel ways. For example, transcatheter access is used for mitral valve repair (TMVR) (Denti et al., 2021). It is also part of a recent

breakthrough at the Vanderbilt Institute for Surgery and Engineering led by Nabil Simaan, who has innovated catheter technology once again to have extraordinary sensory capabilities during procedures to remove blood clots in the brain (Johnson, 2024).

The experience of HM's UI Partnership with Medtronic has inspired the development of new teaching and research programs at HM to develop the convergence of robotics, technologies, and imaging to advance surgical techniques and improve patient outcomes. These programs have grown outward from TAVR/structural heart disease intervention to encompass a range of medical subspecialties. One such initiative is called the Houston Methodist Bookout Center. The Bookout Center is an innovative, hands-on, clinical training facility for healthcare professionals seeking to maintain excellent procedural skills and acquire new proficiencies. This center creates access to world-class resources for research and training under the same roof, which enables the logical, progressive development of collaborative research and training efforts. Another example is the Center for Rapid Device Translation, which supports scientists and inventors through safety studies and regulatory approval to bring their devices to market. Staff within the center provide faculty with the expertise and core facilities needed to validate their ideas and inventions through preclinical studies and open doors for early-phase clinical testing.

These are just two examples of the type of specialty infrastructure that an "atypical organizational vehicle," like an academic health system, is uniquely poised to create, build, implement, maintain, and bring to UI partnerships. Academic health systems alleviate a portion of the heavy regulatory burden physician-scientists face by supporting them through administrative and regulatory responsibilities, offering IP protection for ideas, and giving back valuable time to focus on solving the problems they have identified.

HM and academic health systems are a unique solution at the intersection of thinkers, innovators, doers, and educators. They bring a patient-centered mindset to the equation. Importantly, they are also uniquely positioned to facilitate UI partnerships that focus on AMI and reduce the burden of disease. Without academic health systems, we would lose the culture of innovation they nurture.

To solve medical challenges, physician-scientists must be able to identify problems, find partners, test possible solutions, and report back on whether they work. They help to answer the question: "Will this solution work in a real-world setting?" Academic health systems value and invest in training physician-scientists in both how to innovate and how to keep up with the innovations in their field. Such centers have the infrastructure necessary to validate and scale an innovation in diverse patient settings and healthcare delivery models and to test the ability to disseminate an innovation more broadly. For as long as academic health systems exist, they can continue to serve as facilitating UI partners and foster the fundamental qualities that are so vital to the cycle of medical translation that drives effective AMI. Together, academic health systems and their UI partners will work to align and balance partner priorities, solve the toughest challenges in medicine, reduce the burden of disease, and improve the prospect of hope for patients.

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Transcatheter Heart Valve Technology: Houston Methodist and Medtronic Partnership

Opportunity

Aortic stenosis = narrowing of the aortic valve, common with age

Only treated surgically, innovations to reduce surgical invasiveness led to transcatheter-guided placement of aortic valve replacement (TAVR)

Approach

Partnership between Medtronic (Industry), Houston Methodist (Academic Health System), and several additional CoreValve Clinical Investigators (Universities/AMCs) to design and test TAVR

CoreValve materials and funding provided by Medtronic; clinical trials led by Houston Methodist and others

Surgeon and surgical team training

Execution

Tested first in most severe cases with no other options

Iterative innovations and testing in intermediate and low-risk cases

Outpaced surgical (*i.e.*, open-heart) AVR in 2019

Dissemination to six hospitals within the HM Valve Clinics and training/uptake beyond main hospital

Now available in all 50 states

Outcomes

Decreased risks of mortality, complications, and readmissions

Shorter hospital stays and recovery times

Continuous improvement of the technology through multiple iterations of innovation:

Expanded application in more populations

Advances to approach other valves in the heart

New sensing technology in catheters

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Michelle C. Swick is a Research Development Associate at Houston Methodist Research Institute (HMRI). She specializes in academic research development, particularly for clinical and translational research. She earned a PhD from Baylor College of Medicine in 2011 and has 12 years of research experience in cell and molecular biology, microbiology, and molecular genetics. After completing a postdoctoral research fellowship at UTHealth, she transitioned to focus her career on scientific communication, following her passion to show scientists how to communicate their research effectively to all audiences. She has 10 years of scientific writing and editing experience,

including teaching scientific writing to junior academic faculty and postdoctoral fellows in the biomedical sciences. She has an Editor of Life Sciences (ELS) certification from the Board of Editors in the Life Sciences.

Edward A. Jones is the President and CEO of Houston Methodist Research Institute (HMRI) and the Chief Business Officer of the Houston Methodist Academic Institute. Under his leadership, HMRI became a top 20 independent research institute within its first decade of existence and now has research expenditures exceeding \$300 million annually. He has overseen the development of governance, finance, compliance, and facilities while integrating research with clinical practice. Previously, Jones led multi-site biomedical research operations at Alfred I. duPont Hospital for Children/Nemours Children's Clinic. He earned natural science and advanced business degrees from the University of North Florida. He has served on advisory boards for MedStar Health Research Institute and the University of Texas Medical Branch (Galveston). He currently serves as an advisory member of the Gulf Coast Consortia and as a member of the Houston Health Museum Board of Directors.